OLYMPIC SPORTS PERFORMANCE ANALYSIS

DOCUMENT 04

Results and Conclusion

Summary, interpretation framework, limitations, and future work

Academic Project Documentation

Purpose and Scope

This document records the project's approach based on the project context available. Compare all descriptions and counts with the final Tableau workbook and dataset before academic submission.

Data note: Important: project context describes the workbook as using demonstration/synthetic data. Treat findings as findings from this dataset, not verified historical Olympic facts, unless independently verified.

1. Summary of the Project

The project combines a Tableau workbook with a Flask website to make sports-performance views accessible through a public web link. Its focus includes medal results, athlete participation, annual patterns, regional comparisons, sport categories, host-country references, and an illustrative investment view.

2. Findings Framework

The workbook is designed to help viewers identify differences in medal distribution, changes across years, and variation by region or sport category. Exact ranked findings, peak years, leading countries, and numerical conclusions should be added only after checking the final charts and underlying data. This report intentionally does not invent specific winners or trends.

3. What the Visualizations Help Explore

- Which countries have larger or smaller medal totals in the included dataset.
- Whether medal totals rise, fall, or fluctuate over the represented years.
- How participation counts compare with medal outcomes.
- How regional and sport-category metrics differ.
- How host-country values compare with a selected reference.
- Whether investment and medal measures move together in the demonstration data.

4. Conclusions

A combined dashboard and website can make the project's data easier to access and explore. Tableau supports interactive visual comparison, while Flask provides navigation for the home, dashboard, and story pages. Conclusions depend on transparent data sourcing, validated calculations, and clear metric definitions.

5. Limitations

- The dataset's provenance and whether it is official or synthetic must be stated clearly.
- Approximate record counts should be checked against the final workbook.
- Aggregated measures can hide differences between events, athletes, and sports.
- Investment-versus-medal comparisons cannot establish causality on their own.
- Embedded Tableau views require correct sharing settings and depend on Tableau availability.

6. Future Improvements

- Add a clear data-source and licensing section.
- Document calculated fields and metric formulas.
- Add reproducible data-preparation steps and quality checks.
- Improve mobile layout and accessibility.
- Add explanatory tooltips where helpful.

- Include verified specific findings after reviewing the final dashboard and data.

7. Final Note for Academic Submission

Before submitting, replace this general findings framework with two to five observations taken directly from the published dashboard. For each observation, record the metric, comparison, time period, and chart that supports it. Confirm that the public website, dashboard, and story pages open successfully.